



Moez Ben Haj Hmida

Associate Professor

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Summary

Generative AI researcher and practitioner with over a decade of academic leadership and startup experience. As a Fulbright Scholar and two-time UC Berkeley Visiting Scholar, I specialize in Retrieval-Augmented Generation (RAG) and Agentic AI, bridging the gap between rigorous academic scholarship and commercial product development.

Education

- Dec 2012 **Faculty of Sciences of Tunis, TN**, PhD in Computer Science
Deployment of Data Mining Methods on Grid Computing. Distinction: Very honorable.
- May 2005 **Faculty of Sciences of Tunis, TN**, M.S. in Computer Science
Algorithm Design for Closed Frequent Itemset Extraction
- June 2002 **Faculty of Sciences of Tunis, TN**, M.Eng. in Computer Science

Experience

- Sept 2013 – present **Associate Professor**, National Engineering School of Tunis (ENIT) – Tunisia
Department of Information & Communication Technologies
- Sept 2022 – Aug 2023 **Visiting Scholar**, EECS, UC Berkeley – California
Departement of Electrical Engineering and Computer Science
- Oct 2019 – Aug 2022 **Chief Operating Officer**, iCompass (startup) – Tunisia
- Apr 2017 – Aug 2017 **Visiting Scholar**, EECS, UC Berkeley – California
Departement of Electrical Engineering and Computer Science
- Sept 2008 – Aug 2013 **Assistant Professor**, Higher Institute of Applied Sciences and Technology of Sousse (ISSATSo) – Tunisia
Department of Computer Science
- Sept 2005 – Aug 2008 **Temporary Assistant Professor**, Faculty of Sciences of Tunis (FST) – Tunisia
Department of Computer Science
- Jan 2004 – July 2004 **Java/J2EE developer**, Marekha – Tunisia
- Sept 2002 – Dec 2003 **System administrator (IBM SP2)**, Faculty of Sciences of Tunis (FST) – Tunisia

Mentoring

- 2013 – present Mentored 2 PhD graduates, 10+ Master's students, and an average of 4 undergraduate researchers annually, leading to several student-co-authored publications.

Teaching

- 2017 – present **Big Data Analytics**
Graduate M.Eng (Year 5) MIndS and Telecom elective
- 2023 – present **Cloud Computing**
Graduate M.Eng (Year 5) MIndS (Industrial Modelisation and Simulation)
- 2023 – present **Natural Language Processing**
Graduate M.S Information Processing and Complexity of Living Systems
- 2017 – present **Virtualization**
Graduate M.Eng (Year 5) Computer Science
- 2017 – present **Internet of Things**
Graduate M.Eng (Year 5) Computer Science
- 2013 – 2022 **Systems Programming**
Undergraduate M.Eng (Years 3 & 4) Computer Science
- 2005 – present **Capstone projects**
Graduate M.Eng Computer Science and other specializations: supervised in average 5 students per year

Projects

- Jan 2019 – Dec 2021 **Artificial Intelligence for Social Good**
- Build Language models for under-represented languages.
 - Misogyny Identification in social media.
 - Hate and Abusive speech identification in social media.
 - Fake news.
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 - Fake news.
- Jan 2018 – Dec 2021 **Sentiment Analysis on Dialectal Arabic**
- Build data sets: social networks scraping.
 - Build classification models: applying ML and DL techniques.
 - Accuracy > 90%
- Jan 2017 – Dec 2018 **Political Elections Forecast**
- Detect politically influential messages based on different aspects (messages content, time, sentiment, and emotion).
 - Use case: 2016 U.S presidential election.
 - Lower prediction error than state of the art work.

- Jan 2016 – Dec 2017 **Scala-based Domain Specific Language for concurrent system modeling**
- Use Scala as a Domain Specific Language to create models by connecting actors and setting parameters.
 - Use the DSL to define actors.
- Jan 2015 – Dec 2016 **Prediction of mobile applications breakout**
- Identified 1768 mobile applications by Web scraping using Scrapy and Python
 - 31 K posts and ~2 M comments extracted from Facebook API and stored in MongoDB
 - User sentiment derived from emoticons and bag of words
 - Features engineered and a data set created
 - 70% of breakouts explained by users' interactions and sentiments

Certificates

- Apr 2025 **Building RAG Agents with LLMs**
- Aug 2025 **Building Agentic AI Applications with LLMs**
- Jan 2025 **Building Transformer-Based Natural Language Processing Applications**
- Oct 2021 **Getting Started with Deep Learning**
- Jan 2020 **Project Management Certification (TSPM).**

Volunteer

- Oct 1997 – July 2006 **Youth Association Science of Tunis (AJST) – Tunis, Tunisia**
- Organized and facilitated interactive educational animations, astronomy training programs, and public astronomy events.
- May 2004 – July 2008 **CLL FST (Free Software Club) – Tunis, Tunisia**
- Co-founded the Free Software Club at the Faculty of Sciences of Tunis to promote the use of free software and raise technological awareness across various disciplines. Organized technical and socio-cultural events to unite a diverse community of students and educators.
- Feb 2017 – Aug 2017 **UC Berkeley Spouses, Partners & Parents Association (BSPPA) – Berkeley, California**
- Developed and maintained the organization's website while mentoring new researcher families to accelerate their integration. Shared personal settlement experiences and curated comprehensive recommendations for Bay Area activities, including university events, local schools, museums, and parks.
- Jan 2024 – present **Tunisian AI Society (TAIS) – Tunis, Tunisia**
- Co-founded the TAIS association in Tunisia and led initiatives to bridge academia and industry by organizing AI training programs, business challenges, and public research events. Additionally, facilitated interactive educational sessions and coordinated online masterclasses featuring AI experts from the Tunisian diaspora.

Awards & Grants

- Selected for the Fulbright Enrichment Program (2023)
- Fulbright Visiting Scholar Fellowship (2022)
- First Place Winner, "IoT Meets AI" Hackathon at Sup'Com (March 1, 2020)
- PhD Fellowship under the France-Tunisia Research Support Initiative (2007-2009)

Interests

Natural Language Processing
Generative AI (RAG, LLMs)
Agentic AI
Distributed Computing

Selected Publications

- Apr 2026 **AI-Assisted Modeling: DSL-Driven AI Interactions**
Smyth, Steven, Busch, Daniel, *Ben HajHmida, Moez*, Lee, Edward A, Steffen, Bernhard
dl.acm.org/doi/pdf/10.1145/3786179.3788326 (Proceedings of the 2026 2nd International Workshop on Low-Code Development and Software Architecture)
- Nov 2025 **RAG and Agentic Assistant: A Combined Approach**
Ben HajHmida, Moez, Lee, Edward A
doi.org/10.1007/978-3-032-07132-3_4 (In Steffen, B. (eds) Bridging the Gap Between AI and Reality. AISoLA 2025. Lecture Notes in Computer Science, vol 16220. Springer, Cham.)
- Aug 2025 **A Fully Neural Tunisian Arabic TTS System**
HajHmida, Moez Ben, Haddad, Hatem, Mabrouk, Aymen Ben El Haj
proceedings.mlr.press/v302/ben-hajhmida26a.html (Proceedings of Machine Learning Research)
- Oct 2024 **Context Engineering for AI-Assisted Programming for Domain-Specific Languages**
Ben HajHmida, Moez, Lee, Edward A
doi.org/10.1007/978-3-032-01377-4_2 (International Conference on Bridging the Gap between AI and Reality)
- Feb 2023 **TunBERT: pretraining BERT for Tunisian dialect understanding**
Haddad, Hatem, Rouhou, Ahmed Cheikh, Messaoudi, Abir, Korched, Abir, Fourati, Chayma, Sellami, Amel, *Ben HajHmida, Moez*, Ghriss, Faten
doi.org/10.1007/s42979-022-01541-y (SN Computer Science)
- Dec 2022 **TuniSER: Toward a Tunisian speech emotion recognition system**
Messaoudi, Abir, Haddad, Hatem, Hmida, Moez Benhaj, Graiet, Mohamed
(Proceedings of the 5th International Conference on Natural Language and Speech Processing (ICNLSP 2022))
- Feb 2022 **Automatic Arabic Poem Generation with GPT-2**
Beheitt, Mohamed El Ghaly, *Ben HajHmida, Moez*
[10.5220/0010847100003116](https://doi.org/10.5220/0010847100003116) (In Proceedings of the 14th International Conference on Agents and Artificial Intelligence (ICAART 2022))
- Jan 2021 **Tunisian Dialectal End-to-end Speech Recognition based on DeepSpeech**
Messaoudi, Abir, Haddad, Hatem, Fourati, Chayma, *Ben HajHmida, Moez*, Mabrouk, Aymen Ben Elhaj, Graiet, Mohamed
[10.1016/j.procs.2021.05.082](https://doi.org/10.1016/j.procs.2021.05.082) (Procedia Computer Science)

May 2020 **A Review of Sentiment Analysis Research in Arabic Language**
Oueslati, Oumaima, Cambria, Erik, *Ben HajHmida, Moez*, Ounelli, Habib
doi.org/10.1016/j.future.2020.05.034 (Future Generation Computer Systems)

May 2020 **Predicting mobile application breakout using sentiment analysis of Facebook posts**
Ben HajHmida, Moez, Oueslati, Oumayma
doi.org/10.1177/0165551520917099 (Journal of Information Science)

Academic Services

Organizing/Program Committee of Conferences and Academic Events

- Scientific Committee member of [the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation \(LREC-COLING\) 2024](#), Torino, Italy
- Scientific Committee member of [the 13th Language Resources and Evaluation Conference \(LREC-22\) 2022](#), Marseille, France
- Program Committee member of [the 19th International Conference on Service-Oriented Computing \(ICSOC-21\) 2021](#), Online
- Program Committee member of [the 18th International Conference on Service-Oriented Computing \(ICSOC-20\) 2020](#), United Arab Emirates
- Program Committee member of [the 16th International Conference on Service-Oriented Computing \(ICSOC-18\) 2018](#), Zhejiang, China
- Program Committee member of [the 15th International Conference on Service-Oriented Computing \(ICSOC-17\) 2017](#), Málaga, Spain
- Program Committee member of [the 14th International Conference on Service-Oriented Computing \(ICSOC-16\) 2016](#), Banff, Alberta, Canada
- Program Committee member of [the 13th International Conference on Service-Oriented Computing \(ICSOC-15\) 2015](#), Goa, India
- Organizing Committee member of [International Conference on Knowledge Discovery and Data Analysis \(KDDA\) 2015](#), ESI, Algiers, Algeria
- Organizing Committee member of [International Workshop on Advanced Information Systems for Enterprises \(IWAISE\) 2014](#), Tunis, Tunisia
- Organizing Committee member of [the 3rd International Conference on the Extraction and Management of Knowledge \(EGC-M\) 2012](#), 2012, Hammamet, Tunisia

Associate Editor

- ACM Transactions on Asian and Low-Resource Language Information Processing ([TALLIP](#))

Reviewer

- Science of Computer Programming, 2026
- ACM Transactions on Asian and Low-Resource Language Information Processing (TALLIP), 2024 –2026
- IEEE Access, 2025-2026
- Journal of Information Science, 2021-2022

Invited Talks and Presentations

- How is AI changing Culture & Art today?, Invited speaker, Fulbright Regional Conference, Tunis, Tunisia (July 2026)
- Embedded System Design via Lingua Franca, Invited Lecture (with Edward Lee), [AI-Assisted Programming lectures, AISoLA](#), Rhodes, Greece (October 2025)
- Large Language Models (LLMs) for African Languages and Dialects, Invited Talk, IndabaX Burundi, Virtual (Burundi) (July 2024)
- NLP Applications in Arabic Languages and Dialects, Invited Talk, 3rd edition of the Mediterranean and African Summer School on Artificial Intelligence, Tunis, Tunisia (June 2024)
- The History of Artificial Intelligence, Invited Talk, Fulbright Visiting Scholar Series, SCIENTIFIC CARIBBEAN FOUNDATION, Virtual (Puerto Rico & Honduras) (April 2024)
- Generative AI powered Chatbots, Invited Talk, [11th IEEE Tunisian Students & Young Professionals](#), Hammamet, Tunisia (December 2023)
- Artificial Intelligence for Conversational Agents, Invited Seminar, Polytechnic University of Puerto Rico (May 2023)
- Big Data Architecture, Hands-on Tutorial, SOGETI, Toulouse France (July 2018)
- Parallel and Distributed Data Mining, Invited Seminar, University of Manouba, Tunisia (November 2013)
- Parallel and Distributed Data Mining, Conférence Internationale Francophone sur l'Extraction et la Gestion des Connaissances (EGC), Hammamet, Tunisia (January 2010)

Skills

Languages: Python, C/C++, Java

ML Frameworks: PyTorch, TensorFlow, HF Transformers, ONNX

Agentic tools: LangChain, LangGraph, LlamaIndex

Infrastructure: Hadoop, Spark, Kubernetes, Docker, Ray, AWS, GCP

Research Areas: Deep Learning, Large Language Models (LLMs), RAG (Retrieval-Augmented Generation)

Languages

English: Fluent

Arabic: Native speaker

French: Fluent